Aditya Reddy

adityareddy400@gmail.com | github.com/Aditya-1020 | aditya-1020.github.io

Education

Manipal University Jaipur — Jaipur, India
B.Tech, Electronics and Communication Engineering — *Aug 2024 – May 2028*

Projects

INT8 Systolic-Array Accelerator | *SystemVerilog, LibreLane, OpenROAD, SKY130* — Jun 2026 – Aug 2026

- Completed full RTL-to-GDSII implementation on SKY130: 758×655 μm^2 die, 27.1K std cell instances. Achieved signoff at **200 MHz** across 9 PVT corners with positive setup/hold slack, **zero DRC/LVS/antenna violations**, and **82% utilization**
- Conducted an implementation comparison of unpipelined vs. pipelined INT8 accelerators, with both designs reaching signoff. Quantified area-power-timing trade-offs, showing that doubling throughput to 200 MHz increased power from 12.7 mW to 69.6 mW with sub-1% die-area overhead
- Developed a post-route ECO flow for timing closure at 82% utilization without die-area expansion. Identified 45 critical endpoints from post-route reports, inserted graded drive-strength buffers with manual placement constraints, eliminated all max-slew violations, and limited worst-case IR drop to 1.53 mV
- Customized the OpenROAD clock-tree synthesis to eliminate artificial register padding, preventing 2,781 unnecessary placements driven by slow SRAM clock pins. Extended the LibreLane Classic flow with custom Python substeps for CTS optimization, ECO insertion, and automated per-corner signoff reporting

Dual-Clock Asynchronous FIFO - Formal Verification | *SystemVerilog, SymbiYosys* — Mar 2026 – May 2026

- Designed a 16-bit, 64-entry dual-clock FIFO using Gray-coded pointers, 2-flop synchronizers, reset synchronization, and an OpenRAM SKY130 SRAM macro
- Formally verified no-overflow, no-underflow, and full/empty mutual exclusion with multiclock BMC, and proved pointer-synchronizer properties using k-induction
- Validated reset, boundary, wraparound, concurrent read/write, and asymmetric clock-ratio behavior with cocotb and Verilator

RISC-V RV32IM Processor Core | *SystemVerilog, Vivado, Verilator* — Jul 2025 – Jan 2026

- Implemented a 5-stage RV32IM pipeline with EX/MEM forwarding and load-use hazard handling, supporting the base ISA and multiply/divide extension
- Built hardware multiply/divide units, including a 32-cycle non-restoring radix-2 divider, with pipeline stalling through a ready handshake
- Added a dynamic branch predictor with BTB, Gshare pattern history table, return-address stack, and performance counters for IPC, stalls, and branch behavior
- Verified the core with 28 SystemVerilog assertions across 8 test suites with a 100% pass rate

Low-Latency Trading Gateway | *C++, FIX Protocol* — Jun 2025 – Jul 2025

- Built a low-latency market-data and order-processing pipeline using lock-free queues, memory pools, CPU pinning, and zero-copy FIX parsing
- Measured **2.8 μs** minimum tick-to-trade latency with parser latency averaging **1.4 μs**

Technical Skills

Physical Design: RTL-to-GDSII, floorplanning, placement, CTS, routing, STA, parasitic extraction, physical verification (DRC/LVS/antenna), PVT corner analysis
EDA & PDKs: LibreLane, OpenROAD, Yosys, Magic, KLayout, OpenRAM, SKY130, Vivado
RTL & Verification: SystemVerilog, Verilog, SVA, formal verification, CDC, FPGA
Programming: C, C++, Python

Activities

Volunteer — 2022 – 2024
Bison Asha School for Special Needs — *India*

- Supported classroom educational activities and organized inclusive events for students with special needs